

Test Results – Task ABCDE

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Date: 23.12.2024

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Program : w2119872_(Task ABCDE).py

1. Input validation testing : Tests for data type and range validations.

Test Case	Input	Expected Result	Actual Result	Status	Comments
Day Input Validation	sdfg	Error: "Integer Required!!"	Error: "Integer Required!!"	Passed	Non-integer input handled correctly
	32	Error: "Out of range - values must be in the range of 1 to 31"	Error: "Out of range - values must be in the range of 1 to 31"	Passed	Out-of-range day value detected.
	15	Proceed to month input prompt	Proceed to month input prompt	Passed	Valid input processed correctly.
Month Input Validation	march	Error: "Integer Required!!"	Error: "Integer Required!!"	Passed	Non-integer input handled correctly.
	13	Error: "Out of range - values must be in the range of 1 to 12"	Error: "Out of range - values must be in the range of 1 to 12"	Passed	Out-of-range month value detected.
	3	Proceed to year input prompt	Proceed to year input prompt	Passed	Valid input processed correctly.
Year Input Validation	march	Error: "Integer Required!!"	Error: "Integer Required!!"	Passed	Non-integer input handled correctly.

	1955	Error: "Out of range - values must be in the range of 2000 to 2024"	Error: "Out of range - values must be in the range of 2000 to 2024"	Passed	Out-of-range year value detected.
	2022	Attempt to open file "traffic_data15032022.csv" or show file not found error	File not found error displayed: "traffic_data15122022.csv File Not Found!"	Passed	Valid input processed correctly.

```
*IDLE Shell 3.12.2*
File Edit Shell Debug Options Window Help
Python 3.12.2 (tags/v3.12.2:6abddd9, Feb 6 2024, 21:26:36) [MSC v.1937 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: E:\SD1 CW1\cw D E\w2119872\w2119872_(Task ABCDE).py
Please enter the day of the survey in the format DD : sdfg
Integer required!!
Please enter the day of the survey in the format DD : 32
Out of range - values must be in the range of 1 to 31
Please enter the day of the survey in the format DD : 15
Please enter the month of the survey in the format MM : march
Integer required!!
Please enter the month of the survey in the format MM : 13
Out of range - values must be in the range of 1 to 12
Please enter the month of the survey in the format MM : 3
Please enter the year of the survey in the format YYYY : march
Integer required!!
Please enter the year of the survey in the format YYYY : 1955
Out of range - values must be in the range of 2000 to 2024
Please enter the year of the survey in the format YYYY : 2022
traffic_data15032022.csv File Not Found!
Please enter the data again.

*****
Do you want to select another data file for a different date? (Y/N) : |
```

2. **Edge case testing** : Demonstrates robust handling of edge cases.

Test Case	Input	Expected Result	Actual Result	Status	Comments
Day : Edge Case Testing	1	Proceed to month input prompt	Proceed to month input prompt	Passed	Minimum valid day handled properly
	31	Proceed to month input prompt	Proceed to month input prompt	Passed	Maximum valid day handled correctly.

Month : Edge Case Testing	1	Proceed to year input prompt	Proceed to year input prompt	Passed	Minimum valid month handled correctly.
	12	Proceed to year input prompt	Proceed to year input prompt	Passed	Maximum valid month handled correctly.
Year : Edge Case Testing	2000	Show file not found error if file does not exist	Show file not found error if file does not exist	Passed	Minimum valid year handled correctly.
	2024	Show file not found error if file does not exist	Show file not found error if file does not exist	Passed	Maximum valid year handled correctly.

```

IDLE Shell 3.12.2*
File Edit Shell Debug Options Window Help
>>> = RESTART: E:\SD1 CW1\cw D E\w2119872\w2119872_(Task ABCDE).py
Please enter the day of the survey in the format DD : sdfg
Integer required!!
Please enter the day of the survey in the format DD : 32
Out of range - values must be in the range of 1 to 31
Please enter the day of the survey in the format DD : 15
Please enter the month of the survey in the format MM : march
Integer required!!
Please enter the month of the survey in the format MM : 13
Out of range - values must be in the range of 1 to 12
Please enter the month of the survey in the format MM : 3
Please enter the year of the survey in the format YYYY : march
Integer required!!
Please enter the year of the survey in the format YYYY : 1955
Out of range - values must be in the range of 2000 to 2024
Please enter the year of the survey in the format YYYY : 2022
traffic_data15032022.csv File Not Found!
Please enter the data again.

*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 1
Please enter the month of the survey in the format MM : 1
Please enter the year of the survey in the format YYYY : 2000
traffic_data01012000.csv File Not Found!
Please enter the data again.

*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 31
Please enter the month of the survey in the format MM : 12
Please enter the year of the survey in the format YYYY : 2024
traffic_data31122024.csv File Not Found!
Please enter the data again.

*****
Do you want to select another data file for a different date? (Y/N) : |
Ln: 42 Col: 70

```

3. Functional testing : Validates correctness of all results against sample data.

Metric	Expected Output	Actual Output	Status
Data file selected	traffic_data15062024.csv	traffic_data15062024.csv	Passed
Total number of vehicles	1037	1037	Passed
Total number of trucks	109	109	Passed
Total number of electric vehicles	368	368	Passed
Total number of two-wheeled vehicles	401	401	Passed

Total number of buses leaving Elm Avenue/Rabbit Road (north)	15	15	Passed
Total number of vehicles through both junctions (no turns)	363	363	Passed
Percentage of total vehicles that are trucks	11%	11%	Passed
Average number of bikes per hour	7	7	Passed
Total number of vehicles over speed limit	205	205	Passed
Total vehicles through Elm Avenue/Rabbit Road junction	494	494	Passed
Total vehicles through Hanley Highway/Westway junction	543	543	Passed
Percentage of vehicles through Elm Avenue that are scooters	10%	10%	Passed
Highest number of vehicles in an hour on Hanley Highway	39	39	Passed
Busiest hour of Hanley Highway	18:00 – 19:00	18:00 – 19:00	Passed
Number of hours of rain	0	0	Passed

```

*IDLE Shell 3.12.2*
File Edit Shell Debug Options Window Help
Please enter the month of the survey in the format MM : 1
Please enter the year of the survey in the format YYYY : 2000
traffic_data01012000.csv File Not Found!
Please enter the data again.

*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 31
Please enter the month of the survey in the format MM : 12
Please enter the year of the survey in the format YYYY : 2024
traffic_data31122024.csv File Not Found!
Please enter the data again.

*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 15
Please enter the month of the survey in the format MM : 6
Please enter the year of the survey in the format YYYY : 2024
*****
Data file selected is traffic_data15062024.csv
*****
The total number of vehicles recorded for this date is : 1037
The total number of trucks recorded for this date is : 109
The total number of electric vehicles for this date is : 368
The total number of two-wheeled vehicles for this date is : 401
The total number of busses leaving Elm Avenue/Rabbit Road heading north is : 15
The total number of vehicles passing through both junctions without turning left or right is : 363
The percentage of total vehicles recorded that are trucks for this date is : 11%
The average number of bikes per hour for this date is : 7
The total number of vehicles recorded as over the speed limit for this date is : 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is : 494
The total number of vehicles recorded through Hanley Highway/Westway junction is : 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is : 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is : 0
Ln: 66 Col: 0

```

4. Boundary testing : Tests for edge cases and large datasets.

- Output of date 16-06-2024

```
*IDLE Shell 3.12.2*
File Edit Shell Debug Options Window Help

The total number of two-wheeled vehicles for this date is : 401
The total number of busses leaving Elm Avenue/Rabbit Road heading north is : 15
The total number of vehicles passing through both junctions without turning left or right is : 363
The percentage of total vehicles recorded that are trucks for this date is : 11%
The average number of bikes per hour for this date is : 7
The total number of vehicles recorded as over the speed limit for this date is : 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is : 494
The total number of vehicles recorded through Hanley Highway/Westway junction is : 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is : 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is : 0

Previous data in the results.txt file cleared.
*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 16
Please enter the month of the survey in the format MM : 6
Please enter the year of the survey in the format YYYY : 2024
*****
Data file selected is traffic_data16062024.csv
*****
The total number of vehicles recorded for this date is : 101
The total number of trucks recorded for this date is : 11
The total number of electric vehicles for this date is : 29
The total number of two-wheeled vehicles for this date is : 29
The total number of busses leaving Elm Avenue/Rabbit Road heading north is : 0
The total number of vehicles passing through both junctions without turning left or right is : 38
The percentage of total vehicles recorded that are trucks for this date is : 11%
The average number of bikes per hour for this date is : 0
The total number of vehicles recorded as over the speed limit for this date is : 20
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is : 52
The total number of vehicles recorded through Hanley Highway/Westway junction is : 49
5% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is : 5
The most vehicles through Hanley Highway/Westway were recorded between 01:00 and 02:00
The number of hours of rain for this date is : 3

Ln: 92 Col: 0
```

- Output of date 21-06-2024

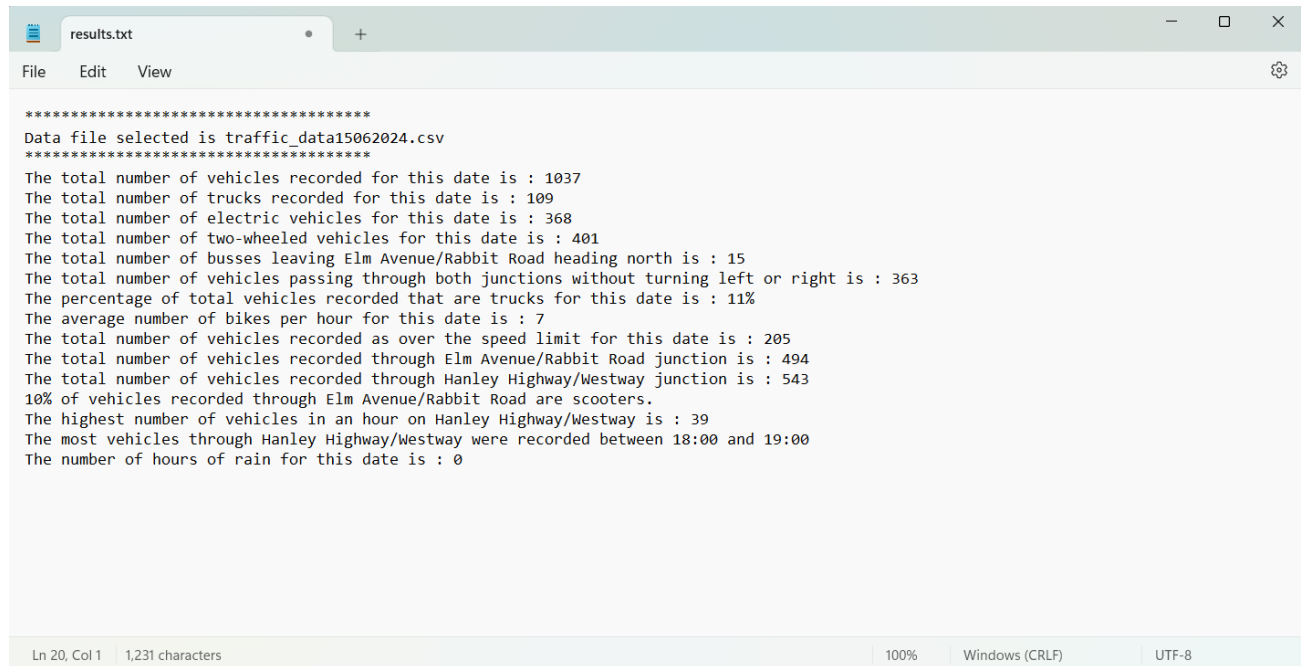
```
*IDLE Shell 3.12.2*
File Edit Shell Debug Options Window Help

The total number of two-wheeled vehicles for this date is : 29
The total number of busses leaving Elm Avenue/Rabbit Road heading north is : 0
The total number of vehicles passing through both junctions without turning left or right is : 38
The percentage of total vehicles recorded that are trucks for this date is : 11%
The average number of bikes per hour for this date is : 0
The total number of vehicles recorded as over the speed limit for this date is : 20
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is : 52
The total number of vehicles recorded through Hanley Highway/Westway junction is : 49
5% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is : 5
The most vehicles through Hanley Highway/Westway were recorded between 01:00 and 02:00
The number of hours of rain for this date is : 3

Previous data in the results.txt file cleared.
*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 21
Please enter the month of the survey in the format MM : 6
Please enter the year of the survey in the format YYYY : 2024
*****
Data file selected is traffic_data21062024.csv
*****
The total number of vehicles recorded for this date is : 1334
The total number of trucks recorded for this date is : 138
The total number of electric vehicles for this date is : 442
The total number of two-wheeled vehicles for this date is : 503
The total number of busses leaving Elm Avenue/Rabbit Road heading north is : 19
The total number of vehicles passing through both junctions without turning left or right is : 494
The percentage of total vehicles recorded that are trucks for this date is : 10%
The average number of bikes per hour for this date is : 10
The total number of vehicles recorded as over the speed limit for this date is : 250
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is : 651
The total number of vehicles recorded through Hanley Highway/Westway junction is : 683
9% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is : 71
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is : 6

Ln: 118 Col: 0
```

5. File output testing : Validates that results are saved correctly.

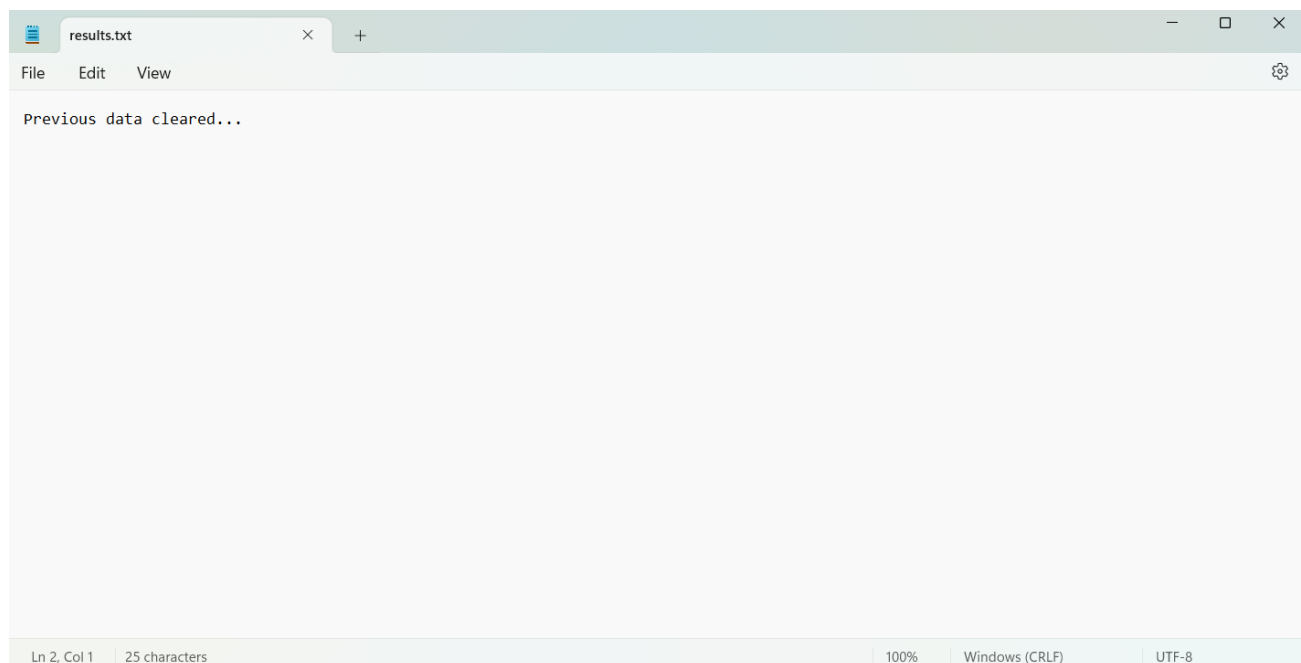


```
*****
Data file selected is traffic_data15062024.csv
*****
The total number of vehicles recorded for this date is : 1037
The total number of trucks recorded for this date is : 109
The total number of electric vehicles for this date is : 368
The total number of two-wheeled vehicles for this date is : 401
The total number of busses leaving Elm Avenue/Rabbit Road heading north is : 15
The total number of vehicles passing through both junctions without turning left or right is : 363
The percentage of total vehicles recorded that are trucks for this date is : 11%
The average number of bikes per hour for this date is : 7
The total number of vehicles recorded as over the speed limit for this date is : 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is : 494
The total number of vehicles recorded through Hanley Highway/Westway junction is : 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is : 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is : 0
```

Ln 20, Col 1 | 1,231 characters | 100% | Windows (CRLF) | UTF-8

6. Append testing : Tests appending functionality for multiple runs.

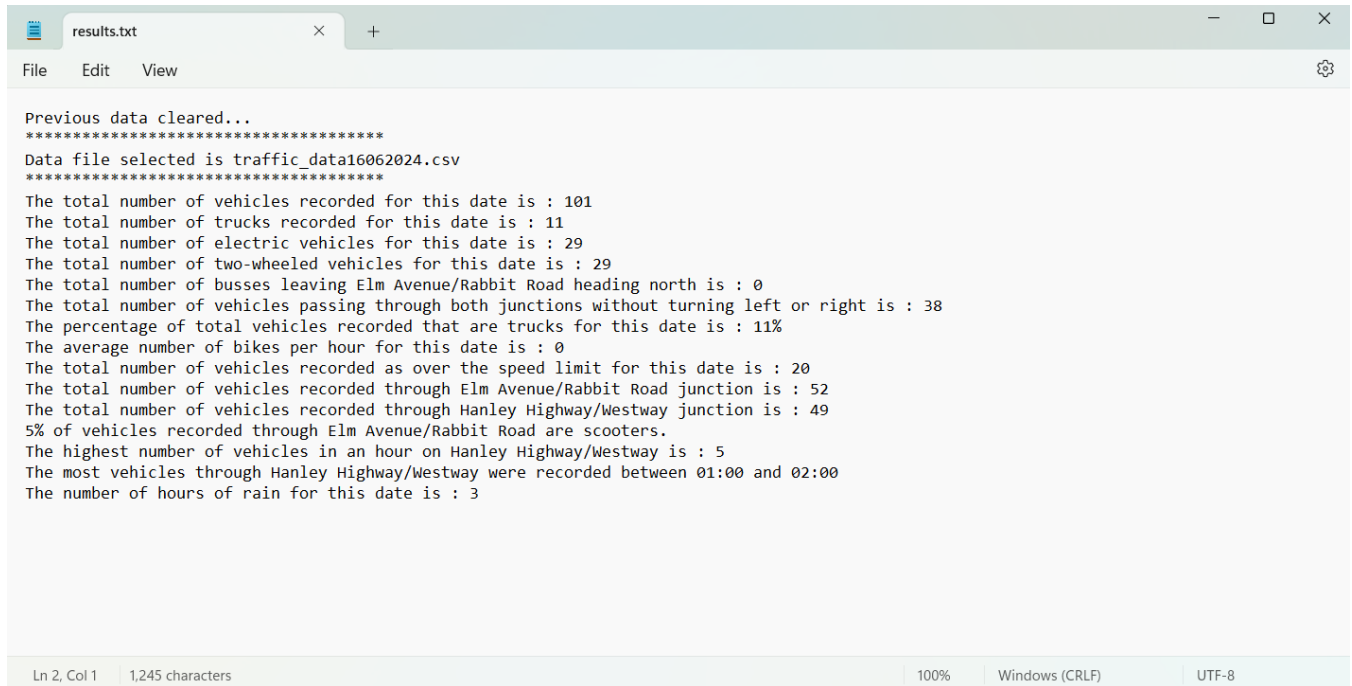
Because of the Task E clear_previous_data method it clears the results.txt file and write over.



```
Previous data cleared...
```

Ln 2, Col 1 | 25 characters | 100% | Windows (CRLF) | UTF-8

The next data set will append to the above file

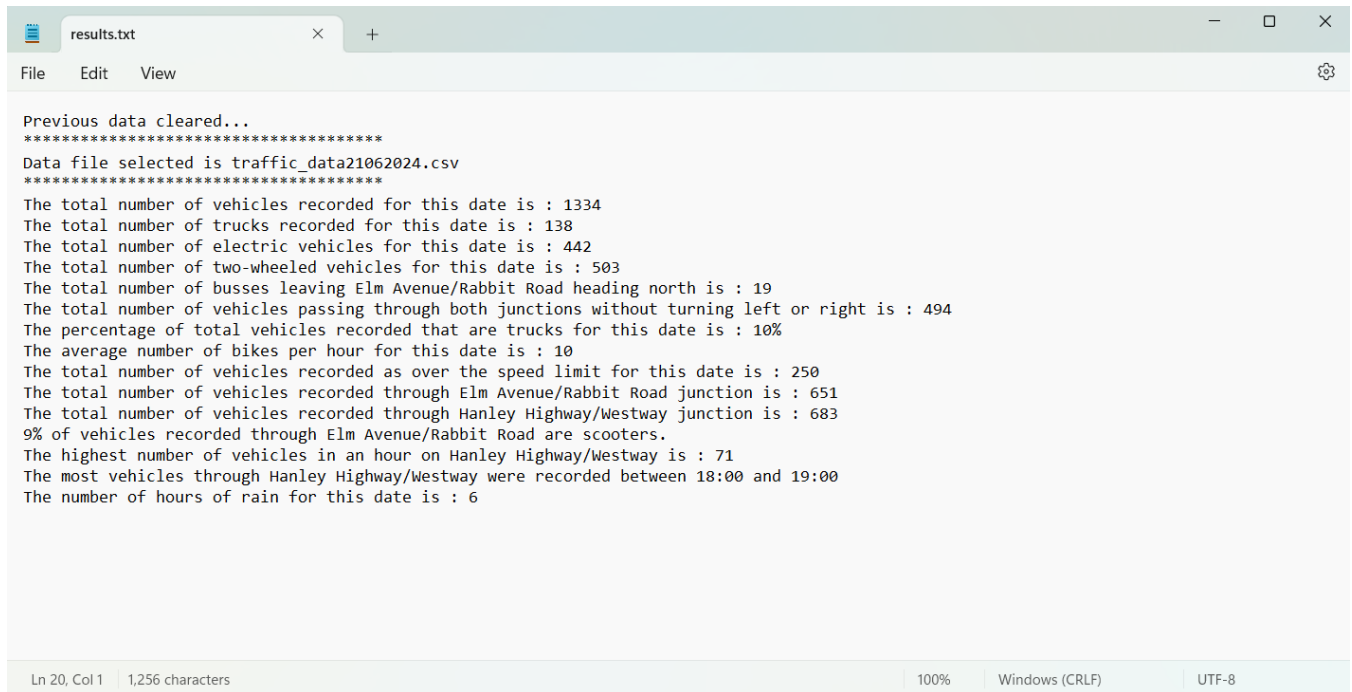


A screenshot of a text editor window titled 'results.txt'. The window has a menu bar with 'File', 'Edit', and 'View'. The text content is as follows:

```
Previous data cleared...
*****
Data file selected is traffic_data16062024.csv
*****
The total number of vehicles recorded for this date is : 101
The total number of trucks recorded for this date is : 11
The total number of electric vehicles for this date is : 29
The total number of two-wheeled vehicles for this date is : 29
The total number of busses leaving Elm Avenue/Rabbit Road heading north is : 0
The total number of vehicles passing through both junctions without turning left or right is : 38
The percentage of total vehicles recorded that are trucks for this date is : 11%
The average number of bikes per hour for this date is : 0
The total number of vehicles recorded as over the speed limit for this date is : 20
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is : 52
The total number of vehicles recorded through Hanley Highway/Westway junction is : 49
5% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is : 5
The most vehicles through Hanley Highway/Westway were recorded between 01:00 and 02:00
The number of hours of rain for this date is : 3
```

The status bar at the bottom shows 'Ln 2, Col 1', '1,245 characters', '100%', 'Windows (CRLF)', and 'UTF-8'.

Data of 21-06-2024 day



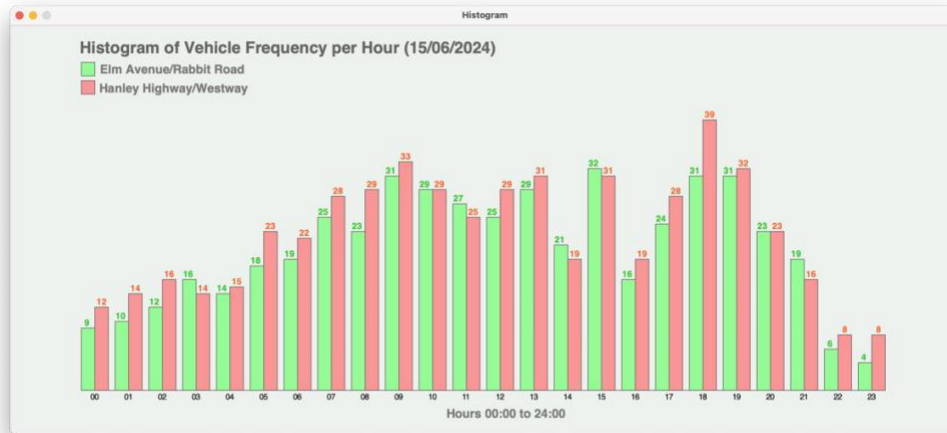
A screenshot of a text editor window titled 'results.txt'. The window has a menu bar with 'File', 'Edit', and 'View'. The text content is as follows:

```
Previous data cleared...
*****
Data file selected is traffic_data21062024.csv
*****
The total number of vehicles recorded for this date is : 1334
The total number of trucks recorded for this date is : 138
The total number of electric vehicles for this date is : 442
The total number of two-wheeled vehicles for this date is : 503
The total number of busses leaving Elm Avenue/Rabbit Road heading north is : 19
The total number of vehicles passing through both junctions without turning left or right is : 494
The percentage of total vehicles recorded that are trucks for this date is : 10%
The average number of bikes per hour for this date is : 10
The total number of vehicles recorded as over the speed limit for this date is : 250
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is : 651
The total number of vehicles recorded through Hanley Highway/Westway junction is : 683
9% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is : 71
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is : 6
```

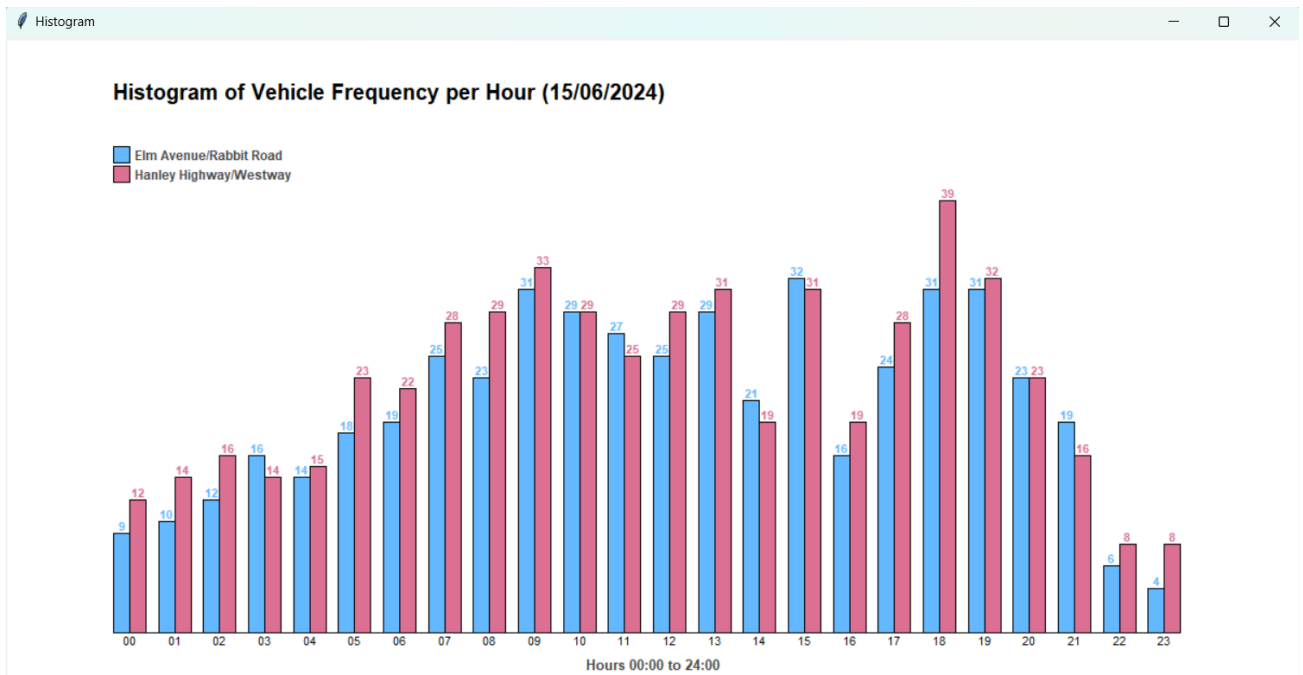
The status bar at the bottom shows 'Ln 20, Col 1', '1,256 characters', '100%', 'Windows (CRLF)', and 'UTF-8'.

7. Graph accuracy testing : Validates graph matches input data.

- Expected result (Date: 15-06-2024)



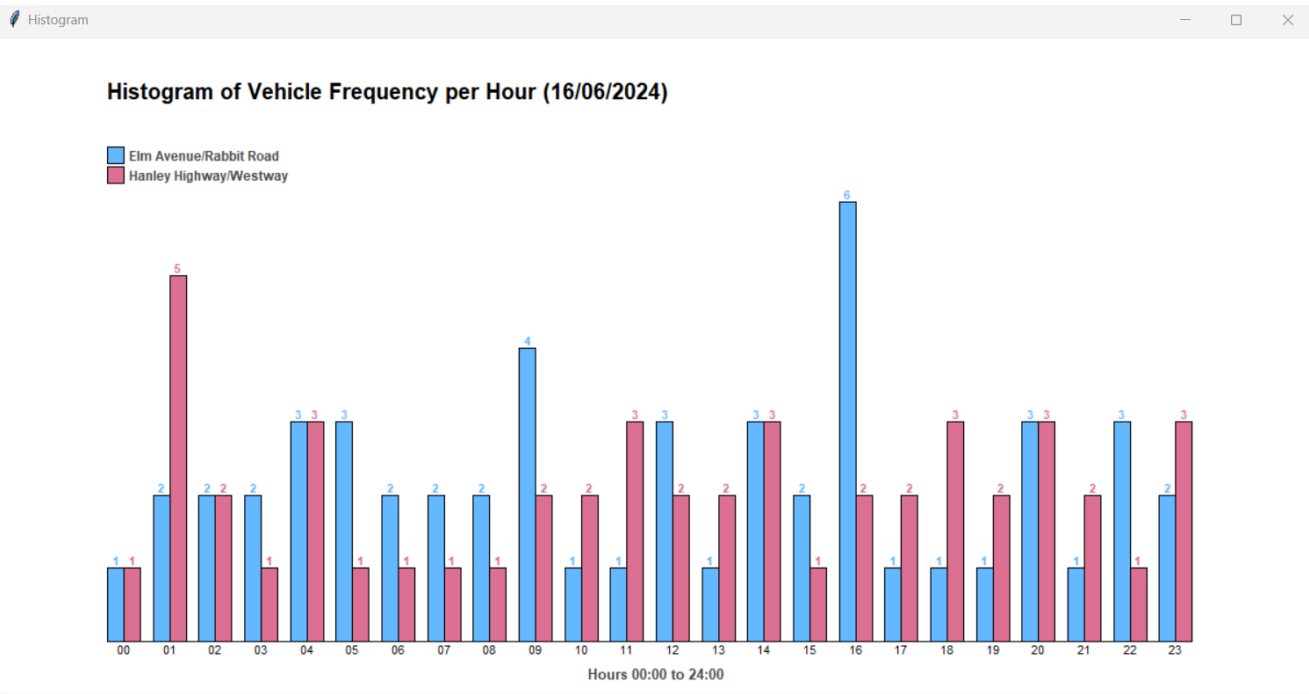
- Actual result (Date: 15-06-2024)



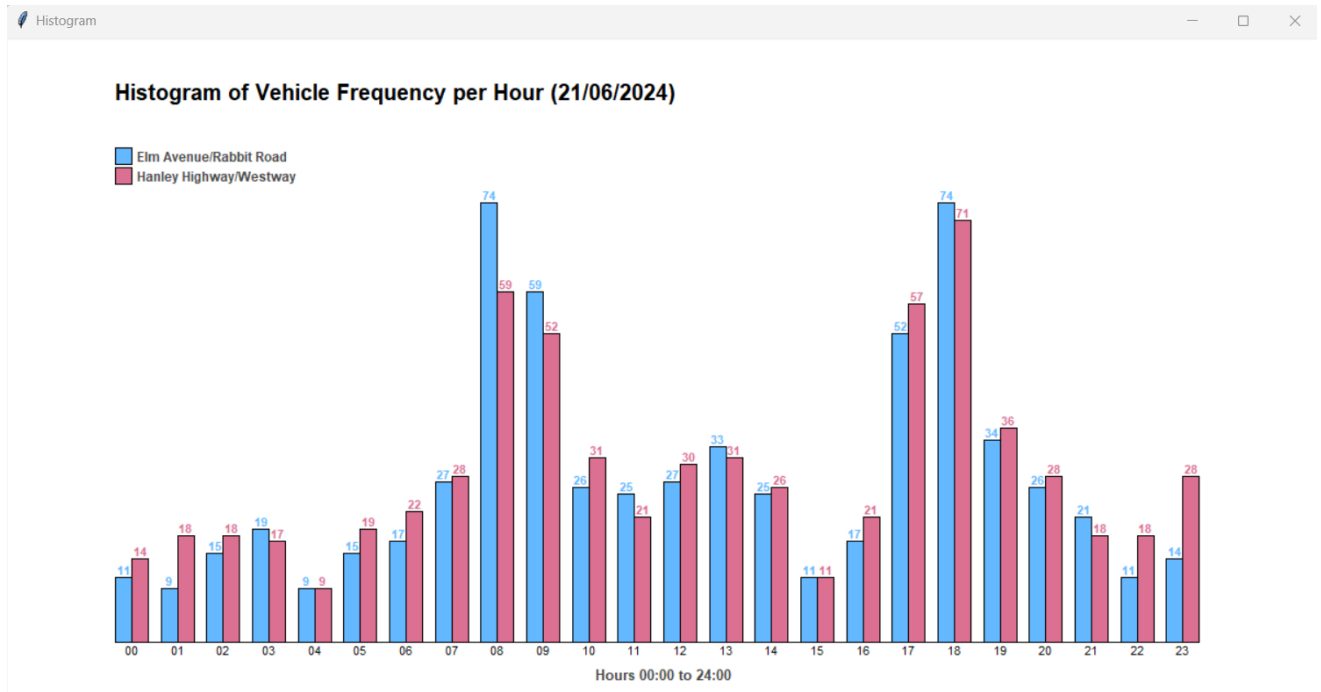
8. **Graph usability testing** : Ensures graph is readable and meets requirements.

Aspect	Criteria	Result (Pass/Fail)
Readability	Graph is easy to interpret, and values are visible.	Pass
Axis Label	X axis is labeled meaningfully according to the given example	Pass
Legends	Legend clearly identifies each data series or category.	Pass
Scaling	Axes are scaled correctly, with even intervals.	Pass
Colours	Colors are distinct and easy to identify.	Pass
Clarity	No overlapping labels, points, or bars.	Pass
Title	Graph has a clear and descriptive title.	Pass

- Histogram of date 16-06-2024



- Histogram of date 21-06-2024



9. **Loop testing** : Validates correct looping through multiple files.

Step 1	Input/Action	Expected Result	Actual Result	Status
Loop through first file	Enter the date as 15-06-2024	Process the traffic_data15062024.csv file and display results.	Process the traffic_data15062024.csv file and display results.	Passed

```

*IDLE Shell 3.12.2*
File Edit Shell Debug Options Window Help
Please enter the month of the survey in the format MM : 01
Please enter the year of the survey in the format YYYY : 2000
traffic_data01012000.csv File Not Found!
Please enter the data again.

*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 31
Please enter the month of the survey in the format MM : 12
Please enter the year of the survey in the format YYYY : 2024
traffic_data31122024.csv File Not Found!
Please enter the data again.

*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 15
Please enter the month of the survey in the format MM : 06
Please enter the year of the survey in the format YYYY : 2024
*****
Data file selected is traffic_data15062024.csv
*****
The total number of vehicles recorded for this date is : 1037
The total number of trucks recorded for this date is : 109
The total number of electric vehicles for this date is : 368
The total number of two-wheeled vehicles for this date is : 401
The total number of busses leaving Elm Avenue/Rabbit Road heading north is : 15
The total number of vehicles passing through both junctions without turning left or right is : 363
The percentage of total vehicles recorded that are trucks for this date is : 11%
The average number of bikes per hour for this date is : 7
The total number of vehicles recorded as over the speed limit for this date is : 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is : 494
The total number of vehicles recorded through Hanley Highway/Westway junction is : 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is : 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is : 0
Ln 359 Col: 0

```

Step 2	Input/Action	Expected Result	Actual Result	Status
Ask from user to continue (y/n)	Enter Y or y	Asks for the date of the file we want	Asks for the date of the file we want	Passed

```

HIDE Shell 3.12.2*
File Edit Shell Debug Options Window Help

*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 31
Please enter the month of the survey in the format MM : 12
Please enter the year of the survey in the format YYYY : 2024
traffic_data31122024.csv File Not Found!
Please enter the data again.

*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 15
Please enter the month of the survey in the format MM : 06
Please enter the year of the survey in the format YYYY : 2024
*****
Data file selected is traffic_data15062024.csv
*****
The total number of vehicles recorded for this date is : 1037
The total number of trucks recorded for this date is : 109
The total number of electric vehicles for this date is : 368
The total number of two-wheeled vehicles for this date is : 401
The total number of busses leaving Elm Avenue/Rabbit Road heading north is : 15
The total number of vehicles passing through both junctions without turning left or right is : 363
The percentage of total vehicles recorded that are trucks for this date is : 11%
The average number of bikes per hour for this date is : 7
The total number of vehicles recorded as over the speed limit for this date is : 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is : 494
The total number of vehicles recorded through Hanley Highway/Westway junction is : 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is : 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is : 0

Previous data in the results.txt file cleared.
*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : |
Ln: 363 Col: 54

```

Step 3	Input/Action	Expected Result	Actual Result	Status
Loop through second file	Enter the date as 16-06-2024	Process the traffic_data16062024.csv file and display corresponding results.	Process the traffic_data16062024.csv file and display corresponding results.	Passed

```

HIDE Shell 3.12.2*
File Edit Shell Debug Options Window Help

The total number of two-wheeled vehicles for this date is : 401
The total number of busses leaving Elm Avenue/Rabbit Road heading north is : 15
The total number of vehicles passing through both junctions without turning left or right is : 363
The percentage of total vehicles recorded that are trucks for this date is : 11%
The average number of bikes per hour for this date is : 7
The total number of vehicles recorded as over the speed limit for this date is : 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is : 494
The total number of vehicles recorded through Hanley Highway/Westway junction is : 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is : 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is : 0

Previous data in the results.txt file cleared.
*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 16
Please enter the month of the survey in the format MM : 6
Please enter the year of the survey in the format YYYY : 2024
*****
Data file selected is traffic_data16062024.csv
*****
The total number of vehicles recorded for this date is : 101
The total number of trucks recorded for this date is : 11
The total number of electric vehicles for this date is : 29
The total number of two-wheeled vehicles for this date is : 29
The total number of busses leaving Elm Avenue/Rabbit Road heading north is : 0
The total number of vehicles passing through both junctions without turning left or right is : 38
The percentage of total vehicles recorded that are trucks for this date is : 11%
The average number of bikes per hour for this date is : 0
The total number of vehicles recorded as over the speed limit for this date is : 20
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is : 52
The total number of vehicles recorded through Hanley Highway/Westway junction is : 49
5% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is : 5
The most vehicles through Hanley Highway/Westway were recorded between 01:00 and 02:00
The number of hours of rain for this date is : 3
|
Ln: 385 Col: 0

```

Step 4	Input/Action	Expected Result	Actual Result	Status
Ask from user to continue (y/n)	Enter N or n	Program terminates gracefully.	Program terminated as expected.	Passed

```

IDLE Shell 3.12.2
File Edit Shell Debug Options Window Help
The average number of bikes per hour for this date is : 7
The total number of vehicles recorded as over the speed limit for this date is : 205
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is : 494
The total number of vehicles recorded through Hanley Highway/Westway junction is : 543
10% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is : 39
The most vehicles through Hanley Highway/Westway were recorded between 18:00 and 19:00
The number of hours of rain for this date is : 0

Previous data in the results.txt file cleared.
*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 16
Please enter the month of the survey in the format MM : 6
Please enter the year of the survey in the format YYYY : 2024
*****
Data file selected is traffic_data16062024.csv
*****
The total number of vehicles recorded for this date is : 101
The total number of trucks recorded for this date is : 11
The total number of electric vehicles for this date is : 29
The total number of two-wheeled vehicles for this date is : 29
The total number of busses leaving Elm Avenue/Rabbit Road heading north is : 0
The total number of vehicles passing through both junctions without turning left or right is : 38
The percentage of total vehicles recorded that are trucks for this date is : 11%
The average number of bikes per hour for this date is : 0
The total number of vehicles recorded as over the speed limit for this date is : 20
The total number of vehicles recorded through Elm Avenue/Rabbit Road junction is : 52
The total number of vehicles recorded through Hanley Highway/Westway junction is : 49
5% of vehicles recorded through Elm Avenue/Rabbit Road are scooters.
The highest number of vehicles in an hour on Hanley Highway/Westway is : 5
The most vehicles through Hanley Highway/Westway were recorded between 01:00 and 02:00
The number of hours of rain for this date is : 3

Previous data in the results.txt file cleared.
*****
Do you want to select another data file for a different date? (Y/N) : n
End of the program.
>>>
Ln: 389 Col: 0

```

10. **Error handling testing** : Ensures errors (e.g., missing files) are handled gracefully.

Handling invalid input during date selection was shown previously.

Test Case	Input	Expected Output	Actual Output	Status
Missing file	Date: 01-02-2024(nonexistent file)	Display “File not found” message and prompt for new input	Error displayed and prompt shown	Passed

```

Previous data in the results.txt file cleared.
*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 01
Please enter the month of the survey in the format MM : 02
Please enter the year of the survey in the format YYYY : 2024
traffic_data01022024.csv File Not Found!
Please enter the data again.

*****
Do you want to select another data file for a different date? (Y/N) :

```

Test Case	Input	Expected Output	Actual Output	Status
Invalid input during "continue loop? (y/n)" selection	Enter anything except y/Y/n/N	Display "Please Enter "Y" or "N"" message and re-prompt the "continue loop? (y/n)" message to user	Error message displayed and re-prompted	Passed

```

*****
Do you want to select another data file for a different date? (Y/N) : yes
Please Enter "Y" or "N"
*****
Do you want to select another data file for a different date? (Y/N) : no
Please Enter "Y" or "N"
*****
Do you want to select another data file for a different date? (Y/N) : 1234
Please Enter "Y" or "N"
*****
Do you want to select another data file for a different date? (Y/N) : |

```

11. Test coverage : Comprehensive test cases for all functionalities.

Functionality	Tested	Comments
Input validation	Yes	Tested for all date components (day, month, year)
Data processing	Yes	Outputs validated against sample data
File handling (valid and invalid files)	Yes	Tested missing files
Performance on large datasets	Yes	Handled large data files without crashing
Creating histogram	Yes	Graph validated against sample graph and graph is readable and meets requirements.
Looping and multiple file processing	Yes	Correctly processes multiple files in a loop
Error handling	Yes	Program recovers gracefully from errors

12. Robustness testing : Ensures program handles unexpected inputs and edge cases.

Test Case	Input	Expected Result	Actual Result	Status
Empty input in date input	Press Enter without typing	Error: "Integer Required!!". Re-prompt user	Error displayed and user re-prompted	Passed

```

*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD :
Integer required!!
Please enter the day of the survey in the format DD : 15
Please enter the month of the survey in the format MM :
Integer required!!
Please enter the month of the survey in the format MM : 6
Please enter the year of the survey in the format YYYY :
Integer required!!
Please enter the year of the survey in the format YYYY : 2024
*****
Data file selected is traffic_data15062024.csv
*****

```

Test Case	Input	Expected Result	Actual Result	Status
Empty input in continue loop input	Press Enter without typing	Error: "Please Enter "Y" or "N"". Re-prompt user	Error message displayed and prompt shown	Passed

```

*****
Do you want to select another data file for a different date? (Y/N) :
Please Enter "Y" or "N"
*****
Do you want to select another data file for a different date? (Y/N) :
Please Enter "Y" or "N"
*****
Do you want to select another data file for a different date? (Y/N) : |

```

Test Case	Input	Expected Result	Actual Result	Status
Edge date (earliest valid date)	Input 01-01-2000 as date	File processed or "File not found" error	File not found error displayed.	Passed

```

*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 01
Please enter the month of the survey in the format MM : 01
Please enter the year of the survey in the format YYYY : 2000
traffic_data01012000.csv File Not Found!
Please enter the data again.

*****
Do you want to select another data file for a different date? (Y/N) : |

```

Test Case	Input	Expected Result	Actual Result	Status
Edge date (latest valid date)	Input 31-12-2024 as date	File processed or "File not found" error.	File not found error displayed.	Passed

```
*****
Do you want to select another data file for a different date? (Y/N) : y
*****
Please enter the day of the survey in the format DD : 31
Please enter the month of the survey in the format MM : 12
Please enter the year of the survey in the format YYYY : 2024
traffic_data31122024.csv File Not Found!
Please enter the data again.

*****
Do you want to select another data file for a different date? (Y/N) : |
```